■ CITY LIBRARY CHATBOT

Sample Chatbot Conversations Report
Rule-Based Chatbot using Python if-else Conditions

Topic	Library Services
Chatbot Type	Rule-Based (Python if-else)
Language	Python 3.x
File Name	library_chatbot.py
Assignment	GMIU Web Portal Submission
Student Name	Dhruvi Pandya

1. Introduction

The **City Library Chatbot** is a rule-based conversational system developed using Python and if-else conditions. It is designed to assist library visitors and members by answering frequently asked questions about library services, book availability, membership, fines, reading rooms, digital resources, and more. The chatbot identifies keywords in user input and returns the most relevant pre-defined response.

2. Chatbot Features

#	Feature	Description
1	Book Search	Guides users to find book availability and catalogue
2	Library Hours	Provides working hours for all days including holidays
3	Membership Info	Explains membership types, fees, and registration steps
4	Book Issue	Explains borrowing process and limits
5	Book Return	Guides return process and drop-box availability
6	Book Renewal	Online, phone, and in-person renewal options
7	Fines & Penalties	Late fee, lost book, and damaged book charges
8	Reservations / Hold	Explains book hold / waitlist system
9	Reading Room	Information about study rooms and 24/7 access
10	Digital Resources	eBooks, e-Journals, NPTEL, WiFi access
11	Contact Info	Address, phone, email, and help desk location
12	Photocopy / Print	Charges and location of copy center
13	Locker Facility	Bag storage and locker rental details

14	Competitive Exams	Reference section and exam book collection
15	Greetings / Exit	Welcome message and polite farewell

3. Sample Chatbot Conversations

The following conversations demonstrate the chatbot's responses to various user queries. Green bubbles represent the User; blue-bordered bubbles represent the Chatbot.

Conversation 1 — Greeting & Services

■ You

Hello

■ Library Bot

Welcome to the City Library Chatbot!
I can help you with:
Book Search & Availability
Library Hours
Membership & Registration
Book Renewal | Fines
Reading Room | Digital Resources
How can I assist you today?

■ You

What services do you offer?

■ Library Bot

Library Services:
Book Lending - Issue & Return books
Reservation - Hold books in advance
Renewal - Extend loan period
Reading Rooms - Silent study spaces
Digital Library - eBooks & Journals
Free WiFi - Throughout library
Photocopy/Print - Rs.1 per page (B&W;)
Competitive Exams - Special collection
Locker Facility - Rs.50/month

Conversation 2 — Library Hours

You

What are the library timings?

Library Bot

Library Working Hours:
Monday - Friday : 9:00 AM - 8:00 PM
Saturday : 10:00 AM - 6:00 PM
Sunday : 11:00 AM - 4:00 PM
Public Holidays : Closed

Note: Reading Room is open 24/7 for registered members.

You

When does the library close on Saturday?

Library Bot

Library Working Hours:
Saturday : 10:00 AM - 6:00 PM
The library closes at 6:00 PM on Saturdays.

Conversation 3 — Book Search & Reservation

You

How can I search for a book?

Library Bot

Book Search & Availability:
1. Visit our online catalogue: www.citylibrary.in/catalogue
2. Use the search terminal at the library counter.
3. Ask the librarian at Help Desk (Counter No. 1).
Search by: Title | Author | ISBN | Subject
If a book is currently issued, you can place a HOLD/RESERVATION.

You

How to reserve a book?

Library Bot

Book Reservation / Hold:
1. Check if book is currently issued.
2. Login to www.citylibrary.in/catalogue
3. Click 'Place Hold' on the book page.
4. You will get an SMS/Email when available.
Hold Period: 2 days | Max holds: 3 books at a time.

Conversation 4 — Membership

You

How can I become a member?

Library Bot

Library Membership:
Types of Membership:
Student : Rs.200/year (5 books)
General : Rs.500/year (7 books)
Senior Citizen: Rs.100/year (7 books)

Documents Required:
ID Proof (Aadhar / College ID)
Passport-size photo (2 copies)
Address Proof
Visit Counter No. 2 to register. Same day processing!

You

How many books can a student member borrow?

Library Bot

A Student Member can borrow up to 5 books at a time.
Loan period: 14 days per book.
Late fine: Rs.2 per day per book after due date.

Conversation 5 — Book Issue & Return

■ You

How do I borrow a book?

■ Library Bot

Issuing a Book:
1. Search for the book in the catalogue.
2. Note the Accession Number / Call Number.
3. Go to the Book Issue Counter (Counter No. 3).
4. Show your Library Card.
5. Book will be issued for 14 days.
Late fine: Rs.2 per day per book after due date.

■ You

How to return a book?

■ Library Bot

Returning a Book:
1. Visit Book Return Counter (Counter No. 3).
2. Hand over the book to the librarian.
3. Collect your return receipt.
Return books within 14 days to avoid fines.
Tip: Self-return drop box near main entrance is also available.

Conversation 6 — Renewal & Fines

You

I want to renew my book

Library Bot

Book Renewal:
1. Online : www.citylibrary.in/renew (Login required)
2. Phone : Call 1800-LIB-RENEW (Mon-Sat)
3. In-Person: Visit Counter No. 3
Rules:
Maximum 2 renewals per book
Each renewal extends loan by 7 days
Renewal not allowed if book is reserved by another member

You

What is the fine for late return?

Library Bot

Fine & Penalty Policy:
Late Return Fine : Rs.2 per book per day
Lost Book Fine : Book price + Rs.100 processing fee
Minor Damage : Rs.50
Major Damage : Full book price
Payment: Cash / Online / UPI at Counter No. 4
Warning: Unpaid fines will block your library account.

Conversation 7 — Digital Resources

You

Do you have ebooks or digital resources?

Library Bot

Digital Resources & E-Library:
Access: www.citylibrary.in/digital

Available Resources:
eBooks - 10,000+ titles
e-Journals - 500+ journals
NPTEL Lectures - Free access
Research Papers- via Shodhganga
DELNET - Inter-library loans
Login: Use your Library Member ID
Free WiFi available inside the library!

Conversation 8 — Contact & Exit

You

What is the library contact number?

Library Bot

Contact Us:
City Library
123, Knowledge Park, Main Road, City - 380001
Landline : 079-2234-5678
Mobile : +91-98765-43210
Email : info@citylibrary.in
Website : www.citylibrary.in
Help Desk: Counter No. 1 (Inside main entrance)
Call us: 9 AM - 8 PM (Mon-Sat)

You

Thank you

Library Bot

Thank you for visiting City Library Chatbot.
Have a wonderful day! Goodbye!

4. How the Chatbot Works

The chatbot is built using a simple but effective rule-based approach in Python. The core logic is handled by the **library_chatbot(user_input)** function. The function follows these steps:

Step 1 — Normalize Input: User input is converted to lowercase and stripped of leading/trailing spaces to make matching case-insensitive and robust.

Step 2 — Keyword Matching: A series of if-elif-else conditions check whether any known keyword is present in the input using Python's 'in' operator and 'any()' function.

Step 3 — Return Response: When a match is found, the corresponding pre-written response is returned. If no match is found, a default help message is shown.

Step 4 — Main Loop: The main() function runs an infinite while loop that continuously reads user input and prints chatbot responses until the user types 'bye' or 'exit'.

5. Keyword Categories

Category	Sample Keywords
Greetings	hi, hello, hey, good morning, namaste
Library Hours	hour, timing, open, close, when, schedule
Book Search	book, search, find, available, author, title
Membership	member, register, join, card, enroll
Book Issue	issue, borrow, take, checkout, lend
Book Return	return, submit, give back
Renewal	renew, renewal, extend, extension
Fines	fine, penalty, late, overdue, fee, charge
Reservation	reserve, hold, waitlist, request
Reading Room	reading room, study, silent zone
Digital	digital, ebook, online, wifi, journal, pdf
Contact	contact, phone, email, address, location
Photocopy	photocopy, print, xerox, copy, scan
Locker	locker, bag, storage, luggage
Farewell	bye, goodbye, exit, quit, thank you

6. Conclusion

The City Library Chatbot successfully demonstrates a rule-based conversational system using Python if-else logic. It covers 15+ library service categories and handles a wide range of user queries through keyword matching. The chatbot is interactive, easy to extend, and provides accurate, helpful responses to

library visitors. It can be further enhanced by integrating a real library database, adding NLP capabilities, or deploying it as a web or mobile application.